

Gathering Validity Evidence for Surgical Simulation

A Systematic Review

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Objective: To identify current trends in the use of validity frameworks in surgical simulation, to provide an overview of the evidence behind the assessment of technical skills in all surgical specialties, and to present recommendations and guidelines for future validity studies.

Summary of Background Data: Validity evidence for assessment tools used in the evaluation of surgical performance is of paramount importance to ensure valid and reliable assessment of skills.

Methods: We systematically reviewed the literature by searching 5 databases (PubMed, EMBASE, Web of Science, PsycINFO, and the Cochrane Library) for studies published from January 1, 2008, to July 10, 2017. We included original studies evaluating simulation-based assessments of health professionals in surgical specialties and extracted data on surgical specialty, simulator modality, participant characteristics, and the validity framework used. Data were synthesized qualitatively.

Results: We identified 498 studies with a total of 18,312 participants. Publications involving validity assessments in surgical simulation more than doubled from 2008 to 2010 (~30 studies/year) to 2014 to 2016 (~70 to 90 studies/year). Only 6.6% of the studies used the recommended contemporary validity framework (Messick). The majority of studies used outdated frameworks such as face validity. Significant differences were identified across surgical specialties. The evaluated assessment tools were mostly inanimate or virtual reality simulation models.

Conclusion: An increasing number of studies have gathered validity evidence for simulation-based assessments in surgical specialties, but the use of outdated frameworks remains common. To address the current practice, this paper presents guidelines on how to use the contemporary validity framework when designing validity studies.

Keywords: Messick framework, surgical simulation, validity framework, validity

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The field of surgical simulation is characterized by innovative approaches that support the acquisition and assessment of surgical technical skills.¹ Simulation models, such as virtual-reality simulators, offer safe environments for the training and evaluation of surgical technical skills without compromising patient safety. As part of continued efforts to establish competency-based training, simulation-based assessments have demonstrated several advantages, including automated feedback and assessments provided by virtual-reality simulators during training.² Currently, the majority of credentialing standards for technical performance in surgery have been established in simulated environments.³ Furthermore, evidence suggests that simulation-based assessments can substitute the often more cumbersome workplace-based assessments, for example, direct observations in the clinic.⁴ The opportunity to perform assessments in a simulated setting has additional potential benefits, including decreased time commitments for faculty members, as well as the ability to obtain objective assessments without the possible influence of observer bias.^{5,6}

However, solid evidence of validity is essential and the use of outdated terms and methods when assessing validity evidence for surgical simulation metrics may lead to unfounded, and even incorrect decisions with respect to physicians' competence.^{7,8} Similar to diagnostic testing and health technology assessments, it is essential to ensure that our tools are assessing the relevant competency/skills. The theoretical basis for the assessment of technical surgical skills has undergone substantial development during the past 2 decades.⁹ Different types of validity (face, content, criterion, and concurrent) have been replaced by a unified model, in which different sources of validity are explored. Messick modern validity framework has become the gold standard when evaluating validity evidence for performance assessments. It was formally adopted by a number of well-recognized associations (American Educational Research Association, American Psychological Association, and National Council on Measurement in Education) in 1999 and the recommendation of this framework was repeated in the most recent edition of Standards for Education and Psychological Testing in 2014.¹⁰ Messick framework consists of 5 different sources of validity evidence: content, response process (ie, quality control), internal structure (eg, reliability), relations with other variables, and consequences of the assessment/test.^{11,12}

In 2010, Korndorffer et al⁸ published a review evaluating the use of validity frameworks in literature concerning laparoscopic education, and found that none of the included studies from 1996 to 2008 used a modern validity framework (Messick) for the investigation of validity evidence.

In this systematic review, we identified trends in the use of validity frameworks in all fields of surgical simulation from 2008 to July 10, 2017. The aims of this study were to provide an overview of the evidence behind the assessment of technical skills in all surgical specialties and to present practical guidelines for future validity studies.

METHODS

Eligibility Criteria

We conducted a systematic review and followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.¹³ We included original studies investigating validity evidence for the assessment of health professionals using surgical simulation in surgical specialties, defined as general surgery (including colon and rectal surgery), urology, orthopedic surgery, obstetrics and gynecology (including gynecologic oncology), vascular surgery, otorhinolaryngology, neurological surgery, ophthalmic surgery, cardiothoracic surgery, pediatric surgery, and plastic and maxillofacial surgery. Studies including surgical tasks across specialties, and studies explicitly including more than one specialty, were also included in the review. We did not limit eligibility based on study design. We only included studies published from 2008 to July 10, 2017. Studies were excluded if they (1) evaluated *nonsurgical* simulation-based assessments; (2) were only descriptive without any attempt to provide evaluation; (3) investigated training or transfer of skills without assessment of validity evidence, or; (4) were published in non-English language.

Search Strategy

We searched 5 databases (PubMed, EMBASE, Web of Science, PsycINFO, and the Cochrane Library) from January 1, 2008, to July 10, 2017, using the following search strategy: (*valid* AND (surger* OR surgeon*) AND (simulation* OR simulator*)*) AND ("2008"[Date - Publication]: "3000"[Date - Publication]). We used Endnote X7.7 (Thomson Reuters, New York, NY) to manage references and eliminate duplicate reports.

Study Selection

One author (NJB) screened all references, removed obviously irrelevant references based on title and abstract, and retrieved full-text of all remaining references. Two authors (NJB and TMHN) then independently read all potentially eligible manuscripts in full-text and assessed study eligibility. Any disagreements were resolved by discussion. If consensus could not be reached, a third author (ASST) made the final decision.

Data Collection, Synthesis, and Analysis

The data extraction was performed independently by 2 authors (NJB and TMHN) based on a pilot-tested data extraction chart. We extracted following data: year of publication, surgical specialty, surgical procedure, simulator modality, participant characteristics, and validity framework used for the assessment of validity evidence. Studies were categorized according to the validity framework that was used: contemporary validity framework (Messick), outdated validity frameworks (eg, criterion, construct, and face validity), or others (no validity framework or concept of validity is defined outside of general scientific norms). For studies applying Messick validity framework, we noted which of the 5 sources of validity evidence were investigated (ie, content, response process, internal structure, relation to other variables, and consequences). For studies applying older validity frameworks, we noted the specific method used and explored transferability to Messick validity framework.

Microsoft Excel 2016 (Microsoft Corp, Redmond, WA) was used to manage the extracted data. Risk of bias within individual studies or across studies was not specifically assessed, as this was not found relevant for the subsequent analysis. It was not possible to conduct a meta-analysis due to the heterogeneity of the included studies. A qualitative data synthesis was performed.

RESULTS

Study Selection and Characteristics

We screened a total of 4600 studies of which 701 were read in full-text. Of those, we excluded 203 studies due to the following main exclusion criteria: only descriptive, not involving simulation, or evaluating nonsurgical skills. Hence, 498 studies were included in the final data synthesis (Fig. 1) (see Supplemental Digital Content for a complete list of references to all of the included studies).

A total of 18,312 participants were included in the 498 included studies (Median 28 participants, range 3 to 344 participants). Specialist surgeons were included in 80.1% of all studies. Other participants were fellows (in 17.7% of studies), residents (in 40.0% of studies), interns (in 5.0% of studies), or medical students (in 27.3% of studies). In 132 studies (26.5%), the participant information provided was insufficient to categorize them, or the participants represented other health care personnel (eg, nurses). Participant characteristics and surgical specialties in review are summarized in Table 1.

Validity Frameworks Used

Publications of validity assessments in surgical simulation more than doubled in numbers from 2008 to 2010 (28 to 35 studies/year, totaling 93 studies) to 2014 to 2016 (68 to 93 studies/year, totaling 233 studies) (Fig. 2). Messick validity framework was used in 6.6% of all studies. No studies used Messick validity framework from 2008 to 2012, but its use has experienced a systematic increase in subsequent years (6.7% in 2013, 10.3% in 2014, 10.8% in 2015, and 12.5% in 2016) with the exception of 2017 (9.3%). We also observed a gradual increase in the frequency of studies not mentioning any validity framework or using incorrect nomenclature from 2.1% in 2012 to 23.3% in 2017.

Use of Messick Contemporary Framework

Of the 33 studies (6.6%) using Messick validity framework, only 5 (15.2%) evaluated all 5 sources of validity evidence (Fig. 3). Internal structure ($n = 25$, 75.8%), content ($n = 22$, 66.7%), and relations to other variables ($n = 21$, 63.6%) were most frequently evaluated, whereas consequences was evaluated least frequently as a source of validity of evidence ($n = 7$, 21.2%) (Fig. 4).

Use of Messick framework differed substantially between different surgical specialties. Its use was most prevalent in cardiothoracic surgery (30.0%) and pediatric surgery (25.8%); less prevalent in ophthalmic surgery (13.3%), plastic and maxillofacial surgery (11.1%), neurosurgery (8.3%), general surgery (7.8%), orthopedic surgery (5.5%), surgery in obstetrics and gynecology (2.8%), and urology (1.2%); and completely absent in otorhinolaryngology and vascular surgery.

Use of Older Frameworks

Overall, 93.4% of the studies used outdated or unspecified validity frameworks. Of these, 80% of the studies had components that were partly transferable to the new framework (construct validity, content validity, reliability, or consequences), while 20% were not directly transferable to the new framework (face validity, criterion validity, or concurrent validity). In 207 studies (41.6% of all studies evaluated), face validity was assessed and considered an acceptable source of validity.

Surgical Specialties

The surgical interventions investigated were categorized according to surgical specialty: Eleven different surgical specialties (classified according to the American College of Surgeons); studies

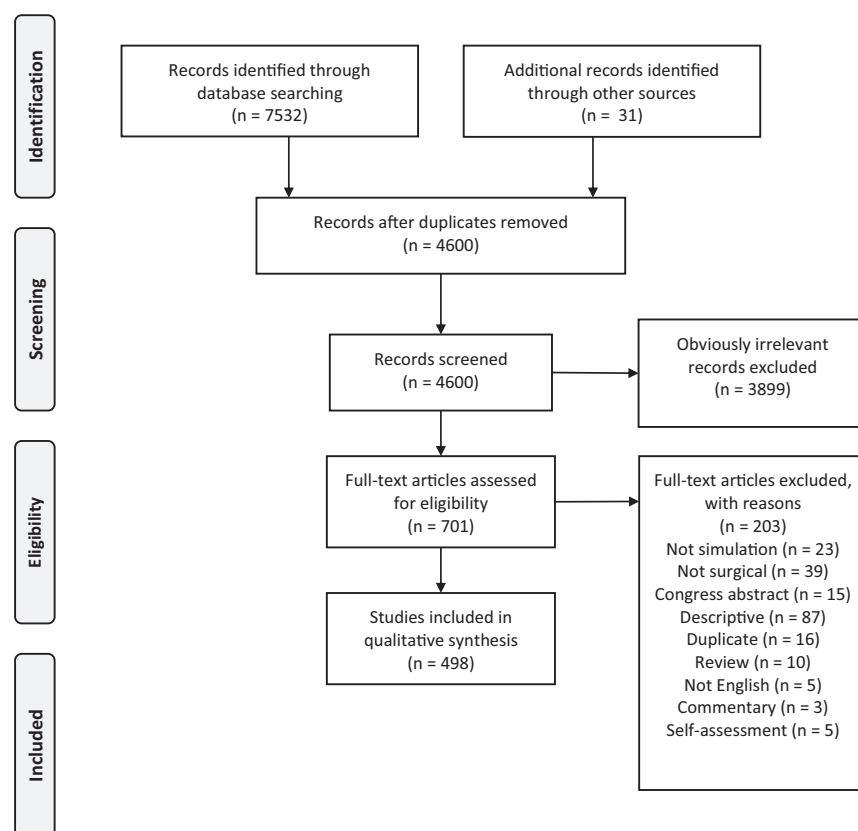


FIGURE 1. PRISMA flow diagram of study selection process.

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

that included tasks across specialties; and studies that explicitly included more than one specialty. The studies were most often performed in the specialties of general surgery (28.3%) and urology (16.9%), while 6.8% of the studies included more than one specialty in their assessment (Table 1).

Simulator Modalities

Simulator modalities were divided into 5 categories: (1) Inanimate models; (2) Animal models; (3) Augmented Reality (AR) models; (4) Virtual Reality (VR) models; and (5) Cadaveric simulation models. Forty-five studies (9.0%) assessed more than one simulator modality making a total of 537 simulator assessments. Inanimate and VR simulation models were used most frequently, 49.9% and 38.7%, respectively. Three studies did not specify the simulation model used. See Fig. 5 for an overview of simulator modality distribution in the included studies.

DISCUSSION

In this systematic review, we evaluated trends in the use of validity frameworks in all fields of surgical simulation from 2008 and onwards. We found that the majority of evidence behind the assessment of surgical technical skills in all surgical specialties is based on outdated validity frameworks. The use of the recommended framework by Messick is increasing slowly, but still more than 9 out of 10 studies (93.4%) rely on outdated methods for the investigation of validity evidence in performance assessments.

Korndorffer et al⁸ found that between 1996 and 2008, all validity studies in laparoscopic simulator education referred to the

TABLE 1. Distribution of Studies Across Specialties and Participant Characteristics

Surgical Specialty	No. of Studies	%
General surgery	141	28.3
Urology	84	16.9
Orthopedic surgery	55	11.0
Otorhinolaryngology	41	8.2
Obstetrics and gynecology	36	7.2
Pediatric surgery	31	6.2
Neurological surgery	24	4.8
Ophthalmic surgery	15	3.0
Vascular surgery	14	2.8
Cardiothoracic surgery	10	2.0
Plastic and maxillofacial surgery	9	1.8
Studies including tasks across specialties	4	0.8
≥2 specialties	34	6.8
Total	498	100
Participant Characteristics	No. of Studies	%
Surgeons	399	80.1
Fellows	88	17.7
Residents	252	40.0
Interns	25	5.0
Medical students	136	27.3
Other healthcare personnel	132	26.5
Not specified	16	3.2
Participants in each study, median (range)	28 (3 to 344)	—
Total number of participants	18,312	—

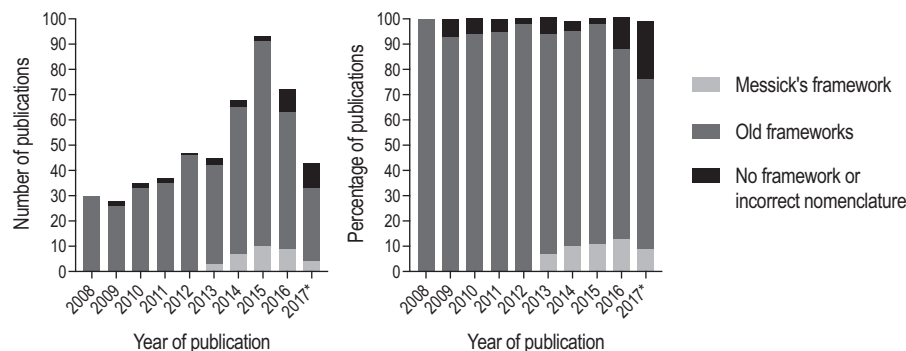


FIGURE 2. Annual number of validity studies in surgical simulation from January 2008 to July 2017. Distribution of used validity frameworks is shown in number of publications (left) and percentage of total publications (right).

old validity framework. Similar results were reported in 2011 by Cook et al¹¹ who found that only 3% of studies involving technology-enhanced simulation used Messick validity framework. According to Ghaderi et al,⁷ Messick's validity framework has only been used for 3 of 23 assessment tools (13%), in the evaluation of procedures/skills that are included in the surgical skills curriculum created by the American College of Surgeons (ACS)/Association of Program Directors in Surgery (APDS). On the basis of a comparison with our current results, we conclude that the use of a contemporary validity framework has improved only slightly. Also, the majority of validity studies investigating the assessment of nontechnical and nonsurgical skills use outdated frameworks.⁹

In 207 (41.6%) of the studies in this review, face validity was identified either as the only type of validity or as a part of the overall validity assessment. Face validity is a subjective evaluation of appearance. It is one of the major threats to modern validity assessment and is an invalid source of validity evidence.¹⁴

Our findings also show that there is a general increase in validity assessments published within all surgical specialties, and that the greatest number of studies have been conducted in either general surgery or urology. Most often, studies have included inanimate or VR simulation models. Of the few studies using the contemporary framework (6.6%), the majority (75.8%) considered 1 to 3 sources, while only 24.2% explored 4 or all 5 sources of validity

evidence. Consequences of the applied assessment/test has been explored least frequently and only in approximately one-fifth of the studies.

A study by Cook et al⁹ published in 2013 reports a trend toward a slight decrease in the number of validity evidence sources investigated after 2008. Especially consequences of the applied assessment/test are rarely investigated, despite the fact that it is a highly relevant source of validity evidence.¹⁵ Goldenberg et al³ identified only 37 studies that established absolute standards for technical performance in surgery. In addition, several of the included studies investigated sources of validity evidence, which are of limited relevance.¹⁶ For example, medical students participated in approximately one-quarter of the studies, even though the performance of medical students rarely has any relevance for the assessment of surgical skills. When evaluating an assessment tool, it is imperative to clearly define the goals of the assessment.¹⁷ If the purpose of the assessment is to discriminate between competent and noncompetent practitioners of a surgical skill, then performing expert-novice comparisons with medical students as the noncompetent participants,

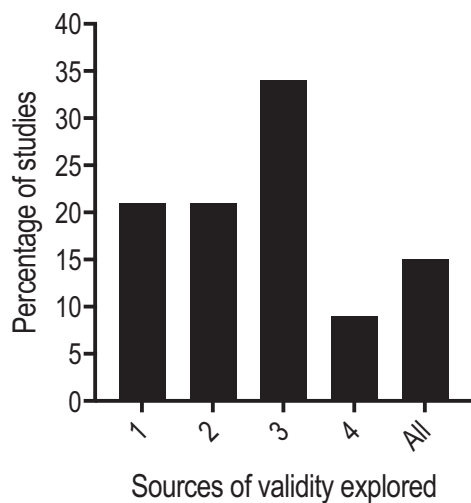


FIGURE 3. Explored sources of validity (within Messick framework) in studies reviewed.

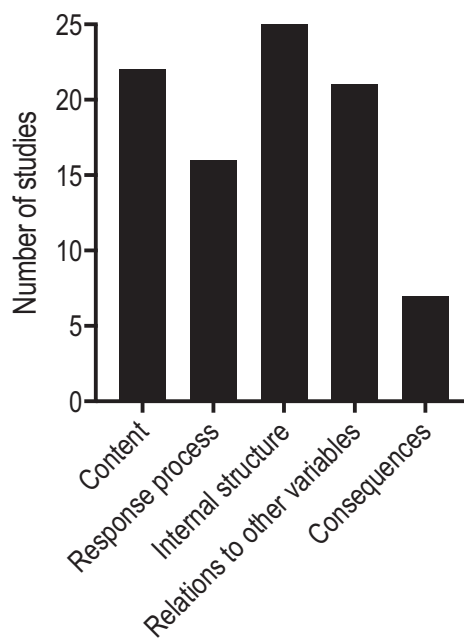


FIGURE 4. Specific sources of validity (within Messick framework) in studies reviewed.

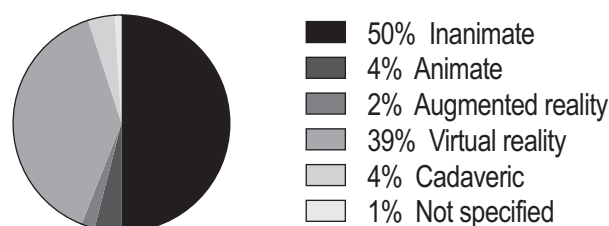


FIGURE 5. Distribution of simulator modalities used in the included studies. In 45 studies, more than one simulator modality was used, and in these cases, all are included (N = 537).

has little relevance, if the intended goal of the assessment is the evaluation of surgical residents.¹⁸ Also, the required reliability coefficient of the assessment should reflect the purpose of the assessment.¹⁹

It is not possible to provide infinite and universal evidence for an assessment tool; on the contrary, an assessment tool is only valid for a specifically defined goal. Thus, collecting validity evidence is a continuous process. On the basis of the results of this systematic review, we would like to emphasize the need for exhaustive and methodologically sound investigations of validity evidence for assessments tools used in surgical specialties. In this paper, we also wish to provide practical guidelines to be used for the design of future validity studies in surgery and as support for the interpretation of existing studies. Table 2 presents an overview of Messick 5 sources of validity evidence, including examples relevant for simulation-based assessments, and serves as practical guidelines for future validity studies. The 5 sources of validity are content, response process, internal structure, relations to other variables, and consequences of the assessment.⁷ Evaluation of *content* must be based on input from

participants who are experienced with the procedure of interest. Data can be collected using questionnaires or by conducting a Delphi study. One example is the development and validation of a theoretical proficiency test for video-assisted thoracoscopic surgery lobectomy, during which experts were interviewed to construct test-items, and subsequently rated these items based on their relevance (a modified Delphi study).²⁰ *Response process* consists of efforts taken to minimize bias in the assessment process, such as standardized instructions and blinded raters. For example, one can achieve this by video recording a surgical procedure of interest and asking raters to assess anonymized videos using a standardized video-solution.^{21,22} *Internal structure* explores the reliability of scores that intend to measure the same construct and can be evaluated statistically using methods, such as Cronbach alpha or generalizability and decision (G- and D-) studies.²³ Reliability statistics can provide insight regarding the applicability of an assessment tool for examinations of different stake levels.¹⁹ *Relations to other variables* include the correlation between assessment scores and external independent measures, such as proficiency level and experience, which have a theoretical relationship to the assessment scores. These analyses can be made using appropriate statistics for correlation (Pearson correlation or Spearman rank-order correlation), and comparisons of 2 groups (independent samples *t* test or Mann-Whitney *U* test) or 3 or more groups (1-way analysis of variance or Kruskal-Wallis test).^{20,21,24} Finally, consequential validity explores potential and actual *consequences* of the developed assessment. Assessment scores stratified by groups of competency can be used to define a standard (eg, using the contrasting groups' method) and explore the consequences of the defined standard on assessment outcomes.^{3,20,21,24} All 5 sources of validity evidence do not necessarily need to be addressed in a single study, and in some cases, not all sources of validity evidence are relevant for a clearly defined purpose. As with all fields of research, the researcher has to critically consider which aspects are relevant for the specific purpose.

TABLE 2. Practical Guidelines for the Use of Messick Validity Framework

Plan	Process	Example	Analyses
Items to be considered	Define goal of assessment	Implement assessment	Conduct appropriate analyses/actions
Content	Ensure that test content reflects the characteristic it intends to measure	Only use expert opinions	Questionnaire Interview Delphi method
Response process	Quality control of the assessment	Minimize bias in your assessment	Standardized written instructions Blinded raters
Internal structure	Reliability of scores across tasks that intend to measure the same construct	Calculate inter-item reliability and/or test-retest reliability. Establish how many items are needed to ensure an appropriate level of reliability. Reflect on the obtained reliability coefficient	Cronbach alpha G- and D-study (generalizability- and decision-study)
Relations to other variables	Correlation between test scores and external independent measures with a theoretical relationship to test scores	Compare scores between groups with different proficiency in the tested skills. Ensure enough participants to achieve normality.	ANOVA Independent sample <i>t</i> test
Consequences of the test	Consequences of the developed assessment	Establish a pass-fail score Determine the consequences of this score.	Contrasting groups method Consequence analysis ROC-analysis (receiver operating characteristic)

Based on Downing and Yudkowsky,¹¹ Downing,¹⁹ and Bloch and Norman.²³

Therefore, we also believe that manuscripts using the old framework still have value, as construct validity, reliability, and content validity are partly transferable to the new framework.

The limitations of this review include the fact that we did not apply specific quality measures to the included studies. Also, we only considered 1 contemporary validity framework (Messick), despite the fact that there are other contemporary validity frameworks, such as Kane's.^{25,26} For this study, we chose to focus only on Messick validity framework, as this remains the most widely accepted contemporary validity framework. We cannot exclude the possibility of publication bias, which is a relevant limitation for most systematic reviews. Furthermore, we omitted interpretation of validity evidence for specific assessment tools. The strengths of this review include the representation of all surgical specialties, as well as all types of simulation models.

Validity studies in surgery demonstrate a significant need for improvement before simulation-based assessments can be systematically integrated for high stakes assessments, including the selection of surgical trainees.²⁷ One reason for slow adoption of a contemporary framework is possibly that many validation studies are conducted by clinicians and, thus, research teams do not necessarily include educational specialists or clinicians with a background in medical education. Clinicians may continue to use outdated frameworks, simply because they are unaware of new developments in this field.

The use of the contemporary validity framework is especially important in the specialties with a limited number of published validity studies, as the early introduction of this framework could have a high impact and influence future publications. By contrast, a gradual shift toward Messick framework in general surgery and urology, which have the greatest number of publications, could prove more difficult. To improve the quality of validity studies, scientific journals should consider adopting a policy that mandates the use of a contemporary validity framework, like Messick. We, therefore, call for a moratorium on the publication of papers containing no or an outdated validity framework. Similar initiatives, such as recommending or requiring the use of the CONSORT statement for randomized trials, have improved the quality of reporting.²⁸ Similar initiatives could help accelerate the widespread use of the contemporary framework.

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